Locale-Conditioned Few-Shot Prompting Prevents Demonstration Regurgitation in On-Device PII Substitution with Small Language Models

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Abstract

PII redaction in machine-learning pipelines today usually replaces detected entities with placeholder tokens such as `[PERSON]`, destroying the downstream utility of the redacted text for retrieval, language-model fine-tuning, and NER training. We propose and evaluate a fully on-device pipeline that *substitutes* PII with consistent, type-preserving fake values: a 1.5 B mixture-of-experts token classifier (`openai/privacy-filter`, with 50 M active parameters) detects spans, an entity resolver groups same-string mentions, and a 1-bit small language model (`Bonsai-1.7B`, Q1_0) proposes contextual surrogates for names, addresses, and dates while a rule-based generator (`faker`) handles fully patterned fields. We are not aware of a prior published evaluation of an end-to-end on-device PII substitution stack that combines a token-classifier, a 1-bit SLM, and a rule-based generator; this paper aims to fill that gap. We report a *prompting* finding that turns out to be more important than the quantization choice: with a naive fixed-three-shot demonstration, both 1-bit `Bonsai-1.7B` and 1.58-bit `Ternary-Bonsai-1.7B` regurgitate the demonstration outputs verbatim regardless of the input—the Chinese name 杨娟, the Japanese name 山田太郎, and the German name Müller-Schulz all collapse to "Alice Johnson". We fix this with **locale-conditioned rotating few-shot demonstrations**: a character-range heuristic classifies each input's locale and selects a locale-pure demonstration pool, and a per-input MD5 hash deterministically samples which three demonstrations appear in the prompt. With the fix, the same 1-bit model produces locale-correct surrogates (杨娟→李伟, Müller-Schulz→Anna Becker, `11-Jul-1998`→`05-Aug-2003`); 482 of 482 unique `Bonsai-1.7B` surrogate-proposal calls succeed (no echoes, no errors) under this strategy. On a generated 2000-document multilingual corpus, hybrid PPL beats `faker` in *all six* locales under a multilingual evaluator (`XGLM-564M`; e.g., zh_CN 68.3 vs. 78.9, ja_JP 55.8 vs. 82.4), and length preservation is the best of three modes in 4 of 6 locales. On a downstream NER experiment (400 train / 100 test, English locales), redact training data yields F1=0.000 (5/5 seeds), faker yields 0.656 ± 0.033, and original-text training yields 0.960 ± 0.004; on a matched 160/40 subset that additionally includes `hybrid`, faker (0.506 ± 0.056) clearly outperforms hybrid (0.346 ± 0.044) at $p < 0.001$ (Welch's $t = 5.0$). We report this as an *honest negative finding*: contextually-realistic SLM surrogates produce more natural-looking text but a less varied training distribution, and downstream NER training benefits more from variety than from naturalness. We argue that small-LM PII substitution is viable on-device, that prompt engineering matters more than quantization at this scale, and that "natural" is not the same as "useful for training." The code repository for this work is accessible at `https://github.com/asadani/on-device-pii-substitution`.

1 Introduction

Existing PII redaction tools—Microsoft Presidio [7], the OpenAI privacy filter [9], `spaCy`+regex pipelines [4]—produce *redacted* text in which detected entities are replaced by short placeholder tokens such as `[NAME]` or `[EMAIL]`. This is adequate for compliance display purposes but fails for downstream uses that need natural-looking text: training data for fine-tuning, search indices over redacted corpora, and retrieval-augmented generation over sensitive document stores. The placeholder approach also hurts utility in measurable ways: NER models trained on `[PERSON]`-redacted corpora generalise poorly to non-redacted text (Section 4.5), and language-model perplexity over placeholder-rich documents is dominated by the placeholder tokens themselves.

Substitution—replacing each detected PII span with a realistic, fake value of the same type—is the obvious alternative, but three constraints have so far prevented its widespread adoption in privacy-sensitive pipelines: **(1)** the substitution must be *consistent* within a document, so that "John Smith" → "Marcus Chen" everywhere, not five different fakes; **(2)** it must be *type-preserving*, so a Chinese name does not become a US name; and **(3)** it should run *on-device* without sending the (still-real-PII) input to a cloud LLM [8].

Recent advances in 1-bit and ternary-bit quantization (e.g., the Bonsai [10] and Ternary-Bonsai [11] families, and prior work on BitNet [6]) make small generative language models viable on commodity CPUs. We investigate whether such models can serve as the surrogate-proposer in a privacy-preserving substitution pipeline, evaluating the resulting system on five metrics (privacy, naturalness, consistency, length, and downstream NER training F1).

Threat model (scope). We measure literal-string leak against ground-truth PII values; we do not consider membership-inference or linkage attacks against the substituted output. Differential-privacy guarantees on the substitution distribution are out of scope.

Contributions.

1. A reproducible on-device substitution pipeline combining `openai/privacy-filter`, `Bonsai-1.7B` (Q1_0), and `faker`, with all code and configurations released.
2. A quantitative comparison of redact / faker-only / hybrid substitution modes under the constraint of CPU-only inference, on 100 multi-template multilingual documents. We are not aware of a directly comparable prior result that fixes all three constraints (consistency, type-preservation, and on-device execution) together.
3. **Identification and isolation of a "few-shot regurgitation" failure mode** in small SLMs at extreme quantization, validated by showing that 1-bit `Bonsai-1.7B` and 1.58-bit `Ternary-Bonsai-1.7B` produce identical regurgitating outputs on the same inputs (i.e., the failure is caused by prompting, not by quantization).
4. **A simple and effective fix: locale-conditioned rotating few-shot demonstrations**, with deterministic per-input demo sampling that preserves cache-friendliness. The fix eliminates the qualitative regurgitation failure (482/482 `Bonsai-1.7B` calls succeed, no echoes) and improves naturalness PPL across all six locales. It does *not*, however, make hybrid surrogates a better source of NER training data than `faker`'s random ones at the larger sample sizes we now evaluate—a tradeoff between naturalness and training-distribution variety that we report as an honest negative finding (Section 4.5).

2 Related Work

2.1 PII detection

Microsoft Presidio [7] combines regex, `spaCy` NER, and rule-based recognisers. The `openai/privacy-filter` model [9] (1.5 B MoE, 50 M active) is a fine-tuned encoder using BIOES / BILOU-style token tags (popularised in NER by Ratinov and Roth [13]) over eight PII categories. Running the inference harness shipped with the model on its bundled seven-template synthetic benchmark (`samples.json`, N=100), we measure F1 = 0.587 (precision 0.619, recall

0.627), with the largest source of false positives being dates (44.5% of all FPs) and the largest source of false negatives being non-English addresses (these numbers are computed by us, from the same harness; see Section 4.1). We use these as the detection-side floor for all subsequent comparisons, since every mode in our experiments inherits the same detector and therefore the same recall ceiling.

2.2 Synthetic PII and anonymization

`faker` [2] is the de-facto rule-based generator for synthetic PII. Presidio's anonymizer can swap detected spans for hash, redaction, or counter-style replacements but not consistent type-preserving fakes. Recent work has used GPT-3.5 / GPT-4 [8] to generate context-appropriate replacements; this approach violates the on-device constraint and incurs per-request cost.

2.3 Low-bit small language models

The Bonsai family (1-bit Q1_0) [10] and Ternary-Bonsai (1.58-bit Q2_0) [11] are Qwen3-based [12] decoder-only LMs trained for extreme quantization. We run inference via `llama.cpp` [3]. Prior work [6] has evaluated extreme-quantization SLMs on QA-style benchmarks; we did not find published evaluations of these models on generative-substitution tasks where input copying or demonstration-regurgitation is a failure mode (the specific failure we document in Section 4.3).

2.4 Downstream-utility evaluation

Prior anonymization work typically reports privacy and naturalness in isolation. We follow the broader synthetic-data literature in additionally measuring length preservation and within-document entity consistency, both relevant to downstream NER training utility, and we evaluate the latter directly by training a `spaCy` [4] NER model on each variant of our corpus and testing on held-out original text.

3 Method

3.1 Architecture

```
text -> privacy-filter (BIOES -> char spans)
     -> EntityResolver (group by canonical, label)
     -> propose_surrogate dispatcher:
         PERSON / ADDRESS / DATE -> Bonsai-1.7B
         EMAIL / PHONE / ACCT / URL / SECRET -> faker
     -> splice (R-to-L, preserve whitespace)
     -> output text
```

Detection runs once per document at $\sim$1 s/512 tokens on CPU. The `EntityResolver` groups detected spans by `(canonical_lowercased, label)` so that all mentions of "John Smith" share a single surrogate. Surrogates are cached per `(mode, family, canonical, label)` so repeated names and common patterns trigger the SLM exactly once over the corpus.

3.2 Surrogate proposers

For PERSON, ADDRESS, and DATE labels, we invoke `Bonsai-1.7B` (Q1_0) through `llama-cli -single-turn`. Each call uses a three-shot prompt of the form `Real: <example>\nFake: <example>\n...Real: <input>\nFake:`.

Locale-conditioned rotating demonstrations. Naively using a small fixed set of demonstrations is catastrophic for 1-bit SLMs: the model pattern-matches the demonstration *outputs* and emits one of them verbatim regardless of the input. A pilot run of the pipeline with three fixed English demonstrations produced “Alice Johnson” as the surrogate for the Chinese name 杨娟, the Japanese name 山田太郎, and the German name Müller-Schulz—all collapsed to the first English demonstration value (see Section 4.3 for the complete diagnosis).

Our final design avoids this with two cooperating mechanisms:

1. **Locale conditioning.** A lightweight character-range and keyword-based heuristic classifies each input into one of {`en`, `de`, `es`, `ja`, `zh`} for PERSON / ADDRESS, and one of {`mdy_slash`, `ymd_dash`, `dmy_dash_mon`, `dmy_slash`, `unknown`} for DATE. Each class has its own pool of 4–8 demonstrations in the matching script and date format (Appendix A).

2. **Per-input rotating sampling.** From the appropriate pool, three demonstrations are sampled deterministically with an MD5 hash of the input string as the random seed. This means *the same entity always receives the same demonstrations* (cache-friendly across documents) but *different entities receive different demonstration subsets* (preventing any single demonstration value from dominating the surrogate distribution).

With this strategy, `Bonsai-1.7B` produces locale-appropriate, format-preserving, and diverse surrogates: 杨娟→李伟, 山田太郎→郑强 (Chinese fallback for kanji-only Japanese names—see Section 5), Müller-Schulz→Anna Becker, `11-Jul-1998`→`05-Aug-2003`.

We validate every response and reject empty, identity-equal, or punctuation-only outputs, falling back to `faker` when validation fails.

For EMAIL, PHONE, ACCT, URL, and SECRET labels, we use `faker` directly because these fields have low contextual entropy—there is no benefit from generative reasoning over `Faker().email()`.

3.3 Splice and whitespace

The privacy filter sometimes includes a leading whitespace in its character span. We preserve original leading and trailing whitespace when splicing surrogates back into the text to avoid cosmetic artefacts such as `is[PRIVATE_PERSON]` (which would inflate perplexity).

4 Experiments

4.1 Setup

Dataset. We use the `openai/privacy-filter` synthetic-document generator [1, 9] (the same template engine that produced the 100-document benchmark shipped with the model card) to generate a fresh, larger 2000-document corpus (`data/samples_2000.json`, seed=42), covering 7 templates (1099, W-2, auto_insurance, bank_statement, invoice, mortgage_insurance, paystub) and 6 locales (en_US n=840, en_IN 320, de_DE 240, es_MX 200, ja_JP 200, zh_CN 200). Each document carries ground-truth PII values ($\sim$6.6 per document on average). Primary metrics (Table 1) are reported on the first 100 documents (en_US 40, en_IN 21, de_DE 12, zh_CN 12, ja_JP 8, es_MX 7); the downstream-NER experiment (Section 4.5) is run on a larger 500-document English subset to give the train/test split enough power for the small-effect comparisons it requires.

Models.

- Detection: `openai/privacy-filter`, 1.5 B MoE, 50 M active, bf16, CPU [9, 14].
- SLM: `Bonsai-1.7B` Q1_0 [10], CPU via `llama.cpp` [3].

Table 1: Primary metrics (N=100, averaged across all 7 templates and 6 locales). Naturalness PPL is computed under the multilingual `XGLM-564M` [5] causal LM (covering all six evaluation locales).

Mode	Leak ↓	PPL ↓	Consistency ↑	Length pres. ↑	Avg. latency
redact	0.249	39.4	1.000	0.979	1.7 s
faker	0.249	84.0	1.000	0.976	1.6 s
hybrid	0.249	**69.9**	1.000	**0.982**	41.2 s

- Naturalness evaluator: `XGLM-564M` (∼564 M parameters, multilingual causal LM trained on 30 languages including all six locales evaluated here) [5], CPU.
- Fallback surrogate generator: `faker` [2].
- Downstream NER: `spaCy` blank English pipeline [4].

Hardware. 31 GB RAM, x86-64 CPU; no GPU used.

Modes evaluated.

- `redact`: replace each span with `[LABEL]` (current state of practice).
- `faker`: replace every span with a `faker` value of matching type.
- `hybrid`: `Bonsai-1.7B` for PERSON / ADDRESS / DATE, `faker` for the other five labels (our proposed pipeline).

4.2 Primary metrics

We report four per-document metrics, averaged across the corpus:

1. **Leak rate**: fraction of *ground-truth PII strings* that still appear verbatim (case-insensitive substring) in the output. Lower is better.
2. **Naturalness PPL**: `XGLM-564M` perplexity over the output text, chunked at 1024 tokens. Lower is more natural. `XGLM-564M` is multilingual and covers all six evaluation locales, so cross-locale PPL numbers are directly comparable rather than biased toward Latin-script outputs.
3. **Consistency rate**: for entities appearing ≥ 2 times in the input, fraction where the output uses the same surrogate at every mention. Higher is better.
4. **Length preservation**: $1 - |\,\mathrm{len(out)} - \mathrm{len(in)}\,|/\mathrm{len(in)}$. Closer to 1 is better.

Headline result. Hybrid PPL (69.9) is lower than faker (84.0, −16.8%) under the multilingual `XGLM-564M` evaluator, and length preservation is the best of the three modes (0.982 vs. 0.976, 0.979). Redact's PPL (39.4) is the lowest of the three numerically, but this is not because the redacted text is more natural—`XGLM-564M` simply scores the `[LABEL]` placeholder as a single out-of-distribution token without further compositional structure to evaluate; we discuss this caveat in Section 5. Consistency is 1.000 across all modes because surrogates are deterministically cached per `(mode, family, canonical, label)`.

Privacy floor. The leak rate of 0.249 is *identical* across all three modes: every mode misses the same ground-truth values, consistent with the detection-side recall (≈ 0.627) we measured for the privacy filter on the same data (Section 2). The architectural choice between redact / faker / hybrid does not affect privacy at all—only *what to do* with the spans the filter catches.

Per-locale (PPL). Hybrid is the XGLM-564M-PPL winner over faker in **all six locales**: en_US (66.1 vs. 80.7, −18.1%), en_IN (74.4 vs. 83.4, −10.8%), de_DE (79.3 vs. 92.5, −14.3%), es_MX (81.1 vs. 101.0, −19.7%), ja_JP (55.8 vs. 82.4, −32.3%), zh_CN (68.3 vs. 78.9, −13.4%). Because XGLM-564M is multilingual, the gap is no longer dominated by tokenisation artefacts, and the ja_JP / zh_CN advantages—where the SLM produces script-correct surrogates that the LM can score in-distribution—are now visible rather than hidden.

Per-locale (length preservation). Hybrid is the length-preservation winner over faker in **four of six locales** (de_DE 0.979 vs. 0.965, en_IN 0.981 vs. 0.972, ja_JP 0.972 vs. 0.964, zh_CN 0.992 vs. 0.974); en_US (0.983 vs. 0.985) and es_MX (0.978 vs. 0.983) narrowly favour faker. The zh_CN gap (+0.018) is the largest of any locale, reflecting the SLM's ability to produce length-matched Han-script addresses where faker's random selection from its zh_CN pool varies more in length.

4.3 Few-shot regurgitation: a prompting failure, not a quantization failure

A pilot run of the pipeline using a fixed three-shot demonstration template (one English, one Japanese, one Spanish demo) revealed a striking failure mode: across all 509 unique-entity `Bonsai-1.7B` calls, the model produced output that was never literally identical to the input (0 echoes by our validation), but for low-resource-locale inputs the output was very often **one of the few-shot demonstration values verbatim, regardless of the input**:

- 杨娟 (Chinese, NAMED INSURED on a `zh_CN` auto-insurance form) → "Alice Johnson" (the first PERSON demonstration).
- 宁夏回族自治区兰州县山亭陈街B座572990 → "123 Main Street, Boston MA 02101" (the first ADDRESS demonstration).
- `11-Jul-1998` (DD-Mon-YYYY) → `03/15/1985` (MM/DD/YYYY—*both* type and locale wrong).
- 山田太郎 (Japanese), Müller-Schulz (German) → both "Alice Johnson".

Diagnosis: prompting, not quantization. Our initial hypothesis was that 1-bit quantization had degraded the model's instruction-following. We tested this by re-running the *exact same five problem prompts* under 1.58-bit `Ternary-Bonsai-1.7B` (Q2_0) [11]—a different quantization scheme running on the same Qwen3 base architecture [12] and the same demonstrations. `Ternary-Bonsai-1.7B` produced **byte-for-byte identical outputs** to `Bonsai-1.7B` (杨娟 → "Alice Johnson", `11-Jul-1998` → `03/15/1985`, etc.) at roughly six times slower per-call inference. This rules out quantization as the root cause: the model is doing what *any* small LM asked to pattern-match `Real:X\nFake:Y` would do—extract the most recent demonstration output as the answer when the input does not match the demonstration's distribution.

Fix: locale-conditioned rotating few-shot demonstrations (described in Section 3.2). With character-range locale detection feeding into pools of 4–8 demonstrations per locale (and 4 demonstrations per date format), and per-input-hash-seeded sampling of 3 demonstrations per call, the same 1-bit `Bonsai-1.7B` produces:

- 杨娟 → 李伟 (Chinese name in zh-pool)
- 宁夏回族自治区兰州县山亭陈街B座572990 → 广东省广州市天河区珠江新城200号 (Chinese address)
- `11-Jul-1998` → `05-Aug-2003` (DD-Mon-YYYY format preserved)
- Müller-Schulz → Anna Becker (German name)

Table 2: Average per-document latency (seconds, CPU only).

Mode	Detect	Surrogate	Splice + PPL	Total
redact	∼1.7	0.00	<0.1	1.7
faker	∼1.6	<0.01	<0.1	1.6
hybrid	∼1.5	∼39.8	<0.1	41.2

- `Hauptstraße 45, 10117 Berlin` → `Bahnhofstraße 7, 60313 Frankfurt`
- "John Smith" → "David Kim" (different US name; no longer "Alice Johnson")

The fix is dataset-free, does not require fine-tuning, and adds < 50 ms per surrogate-proposal call. All quantitative numbers in Tables 1 and 3 are produced under this fixed prompting strategy.

Known limit: kanji-only Japanese names. These map to the `zh` pool because our locale heuristic requires kana to disambiguate (山田太郎→郑强 instead of a Japanese fake). A character-frequency-based classifier or an explicit "treat ambiguous CJK as both ja and zh" strategy would address this.

4.4 Latency

The 24× latency gap between hybrid and redact / faker is dominated by `Bonsai-1.7B` surrogate generation. Without our `(canonical, label)` cache, hybrid would call `Bonsai-1.7B` for every PII mention (∼660 calls across the 100-document corpus); with the cache, only **482 unique calls** were made (482/482 succeeded, 0 echoed-or-empty, 0 errored under the locale-conditioned prompting strategy), saving ∼27% of inference time. Each `Bonsai-1.7B` call costs ∼7–10 s for the ∼30-token output (loading + 1.7 B model inference + cleanup). Detection cost is identical across modes because the privacy filter is invoked both for input PII detection and for residual-leak measurement.

4.5 Downstream NER utility

Setup. We test whether substitution preserves *downstream training utility*: an NER model trained on substituted data should approach the F1 of one trained on original data. From the 2000-document corpus (Section 4.1) we draw the English-locale subset (1159 documents, en_US + en_IN). We run the experiment at two scales: a **large-scale** 400 train / 100 test split for `original`, `redact`, and `faker`, and a **matched-subset** 160 train / 40 test split for all four modes including `hybrid`. The two-scale design lets us report the high-statistical-power comparison between the three cheap modes at the larger scale, while keeping the (CPU-bound) `Bonsai-1.7B` surrogate generation tractable for the four-way comparison at the smaller scale; both scales share the same stratified-by-locale split methodology. The test set is **always in original form**—substitution applies only to the training set. PII spans are extracted from the ground-truth `pii_gt` dictionary (substring search) so that substitution quality is decoupled from detection quality. A blank `spaCy` English NER pipeline is trained for 30 iterations with a single binary `PII` label on each variant of the train set, then evaluated against held-out original spans (label-agnostic span overlap).

Redact destroys downstream utility entirely (F1 = 0.000). A NER model trained on `[PRIVATE_PERSON]`-style placeholders learns to predict only those placeholder *tokens* and never fires on real text. This is the sharpest possible motivation for substitution: redacted-text training

Table 3: Span-level NER F1 on held-out original documents. Mean ± SD across 5 spaCy training seeds; the substituted training corpus is fixed per mode, only spaCy gradient initialisation varies. **Top:** large-scale 400 train / 100 test, three modes. **Bottom:** matched 160 train / 40 test subset including the `hybrid` mode, which requires `Bonsai-1.7B` surrogate generation and is therefore evaluated at the smaller scale to keep wall-clock manageable on CPU.

Mode	Train spans	Precision	Recall	F1	ΔF1 vs. orig.
Large-scale (400 train / 100 test):					
original	2592	0.947 ± 0.006	0.974 ± 0.012	$\mathbf{0.960 \pm 0.004}$	(baseline)
redact	2592	0.000 ± 0.000	0.000 ± 0.000	$\mathbf{0.000 \pm 0.000}$	-0.960
faker	2592	0.984 ± 0.008	0.493 ± 0.038	$\mathbf{0.656 \pm 0.033}$	-0.304
Matched subset (160 train / 40 test):					
original	1046	0.895 ± 0.006	0.923 ± 0.004	$\mathbf{0.908 \pm 0.003}$	(baseline)
redact	1046	0.000 ± 0.000	0.000 ± 0.000	$\mathbf{0.000 \pm 0.000}$	-0.908
faker	1046	0.955 ± 0.017	0.347 ± 0.052	$\mathbf{0.506 \pm 0.056}$	-0.402
hybrid	1046	0.909 ± 0.015	0.215 ± 0.033	$\mathbf{0.346 \pm 0.044}$	-0.562

data is, for any downstream NER consumer, effectively *labelled negative-only data*. The model converges (training loss falls to <2.0) but its decision boundary lives in the wrong vocabulary.

Faker and hybrid both recover a substantial fraction of original F1. Type-preserving fakes—whether random (faker) or SLM-generated locale-conditioned (hybrid)—give the model enough surface variety to learn that "capitalised proper-noun-shaped two-token sequence in a name field" is a PII span. At the large scale (400 train / 100 test), faker recovers $0.656/0.960 = 68.3\%$ of the original baseline; precision is high (0.984 ± 0.008) and recall is moderate (0.493 ± 0.038)—fake-data-trained models are conservative, preferring to miss spans rather than hallucinate them. The seed-to-seed standard deviation on faker F1 is tight at 0.033, putting all between-mode comparisons solidly above the noise floor.

The hybrid–faker comparison at the matched scale: the gap is now real, not noise. On the matched 160 train / 40 test subset—where all four modes share the same train/test docs and all 1046 training PII spans—we observe hybrid F1 $= 0.346 \pm 0.044$ versus faker F1 $= 0.506 \pm 0.056$, a gap of -0.160 in faker's favour. Under Welch's two-sample t-test ($n_{\mathrm{f}} = n_{\mathrm{h}} = 5$, unequal variances) this gives standard error SE $= \sqrt{0.056^2/5 + 0.044^2/5} = 0.032$, $t = 5.02$, Welch–Satterthwaite degrees of freedom ≈ 7.6, and a two-tailed $p < 0.001$. The gap is well outside seed-level noise.

This is an honest negative finding for the central hybrid-vs-faker comparison on *downstream NER training utility*, even though hybrid wins on every other metric we measure: hybrid PPL beats faker under `XGLM-564M` in all six locales (Table 1) and length preservation is best-of-three in four locales. The interpretation is that locale-conditioned, contextually-realistic surrogates produce more natural-looking output text but a *less varied* surface distribution than faker's i.i.d. random surrogates—and downstream NER training benefits more from variety than from naturalness. We revisit this tradeoff in Section 5 and leave open the question of whether a less greedy SLM sampling strategy (higher temperature, or stochastic demonstration sampling without deterministic hash seeding) would close the gap.

A pilot run with the naive fixed-three-demonstration strategy (Section 4.3, single seed) produced hybrid F1 $= 0.310$ vs. faker F1 $= 0.427$—a gap of -0.117. The matched 160/40 gap is -0.160, which is *larger*, not smaller, than that pilot. We read this as: locale-conditioned prompting eliminates the qualitative demonstration-regurgitation failure (482/482 unique `Bonsai-1.7B` calls succeed; Section 4.3), but the resulting in-distribution surrogates still yield less downstream training variety than faker's random ones. Prompting fixes the *output* of the SLM; it does not turn that SLM's output into a better training-data generator than `faker` for this particular task.

5 Discussion and Limitations

Privacy-filter recall ceiling. The non-zero leak rate in all three modes reflects the privacy filter's measured detection F1 of 0.587 (Section 2). A pipeline using a higher-recall detector or pattern-based fallback (e.g., regex for SSNs and pre-masked digits) would shift this floor for all modes equally; the relative comparison between substitution modes is unaffected.

Few-shot regurgitation is a prompting issue, not a quantization issue. Section 4.3 shows that the failure occurs identically under 1-bit `Bonsai-1.7B` and 1.58-bit `Ternary-Bonsai-1.7B`. Larger or higher-precision models *might* suppress the symptom by attending more strongly to the instruction, but the underlying pattern-matching pull will remain at any small-LM scale. The robust fix is at the prompt level (locale-conditioned rotating demonstrations); we recommend it as default practice for any small-LM substitution pipeline.

No coreference resolution. "John" and "John Smith" within the same document are treated as distinct entities by surface-form grouping. A proper coreference layer would further improve consistency.

Naturalness via `XGLM-564M` PPL. We use `XGLM-564M` [5], a 564 M-parameter multilingual causal LM covering all six evaluation locales, as the naturalness proxy. PPL under a generic LM remains a proxy: it rewards in-distribution surface form rather than literal naturalness, and it rewards short placeholder tokens (which `redact` mode produces) for being predictable as standalone tokens regardless of whether the surrounding text is grammatical. A human evaluation, or a more discriminative proxy such as a fluency classifier, would refine this measurement.

No explicit threat model. We measure literal-string leak, not inference attacks. A determined adversary with access to (output_text, external knowledge) could potentially link surrogates back to originals. Differential-privacy guarantees on the substitution distribution are out of scope.

Sample size and statistical reporting. The downstream-NER experiment is run at two scales: a large-scale 400 train / 100 test split for `original`, `redact`, and `faker`, and a matched 160 train / 40 test split that additionally includes `hybrid`. We report mean ± SD per mode across $n = 5$ training seeds and apply Welch's t-test to the substituted-mode comparisons (Table 3). All pairwise comparisons are multiple SDs apart and resolve at $p < 0.001$, including the `hybrid` vs. `faker` comparison at the matched scale (Section 4.5).

6 Conclusion

We built and evaluated a fully on-device PII substitution pipeline combining a token classifier (`openai/privacy-filter`), a 1-bit small language model (`Bonsai-1.7B`, Q1_0), and a rule-based generator (`faker`). The architecture works as intended at the document level: privacy is preserved (within the privacy-filter's recall ceiling), surrogates are consistent across mentions, length preservation is best-of-three in 4 of 6 locales, and naturalness PPL under a multilingual evaluator beats faker-only baselines across *all six* locales.

The most actionable contribution is methodological. We document a "few-shot regurgitation" failure mode in which a small SLM, prompted naively with a fixed three-shot demonstration template, ignores the input and emits one of the demonstration *outputs* verbatim—and we show by direct comparison with 1.58-bit `Ternary-Bonsai-1.7B` that this is a property of *prompting* at small-LM scale, not of 1-bit quantization. A simple locale-conditioned rotating-demonstrations prompting strategy fixes it: 482/482 unique `Bonsai-1.7B` surrogate proposals succeed (no echoes, no errors) under this strategy, producing locale-correct, format-preserving surrogates (杨娟 → 李伟, Müller-Schulz → Anna Becker, `11-Jul-1998` → `05-Aug-2003`); the fix adds < 50 ms per surrogate proposal.

Substitution outperforms `[LABEL]`-style redaction by an enormous margin on downstream NER training utility: at the large 400 train / 100 test scale, faker recovers F1=0.656 ± 0.033 against redact's 0.000 ± 0.000 (a gap of many standard deviations, not a noise effect).

Our negative finding for hybrid is the second methodological contribution. At the matched 160/40 scale, faker (F1=0.506) clearly beats hybrid (F1=0.346) at $p < 0.001$, despite hybrid winning every other metric (PPL, length preservation, locale-fidelity of generated strings). The implication: *natural-looking* substitution and *useful-for-training* substitution are different objectives. SLM surrogates collapse to a narrower in-distribution surface than faker's i.i.d. random outputs, and downstream NER training benefits more from the variety than from the naturalness. We caution future work against treating "hybrid beats faker on PPL" as a proxy for downstream utility—they are not the same metric.

All code, configurations, and the 2000-document generated evaluation corpus are released under an open license.

Reproducibility

All code and the 2000-document generated evaluation set (`data/samples_2000.json`, produced by the `openai/privacy-filter` template engine at `seed=42`) are released at:

`https://github.com/asadani/on-device-pii-substitution`

To reproduce the primary results:

```
# Primary metrics (Table 1, ~3 hours on CPU)
python eval_substitution.py --n 100 --modes redact faker hybrid \
       --samples-path data/samples_2000.json \
       --bonsai-size 1.7B --output results/run_v3

# Per-locale and per-template breakdowns
python analyze_results.py \
       --results results/run_v3/substitution_results.json \
       --out results/run_v3/analysis.md

# Downstream NER utility -- large scale, 3 modes (Table 2 top, ~2.5 h)
python eval_ner_utility.py \
       --samples-path data/samples_2000.json \
       --modes original redact faker --n-max 500 \
       --output results/ner_v3

# Downstream NER utility -- matched 4-mode subset (Table 2 bottom, ~1.5 h)
python eval_ner_utility.py \
       --samples-path data/samples_2000.json \
       --modes original redact faker hybrid --n-max 200 \
       --output results/ner_v3_matched
```

Total wall-clock for the full pipeline is approximately 7–8 hours on a 31 GB single-CPU machine, dominated by hybrid-mode `Bonsai-1.7B` generation in both the primary and NER experiments. The privacy-filter model ($\sim$2.8 GB) and the `XGLM-564M` evaluator ($\sim$1.1 GB) are downloaded once from Hugging Face on first run and cached locally.

Use of generative AI assistance

In line with arXiv's policy on the use of generative AI in scholarly work, the author discloses that a large language model assistant (Anthropic's Claude) was used during the preparation of this manuscript. Specifically: (i) the assistant helped explore and draft prose for the introduction, related-work, and discussion sections from the author's notes and bullet-point outlines; (ii) the assistant helped scaffold LaTeX, bibliography, and table formatting; and (iii) the assistant ran the experimental scripts and reported the resulting numerical outputs back to the author. All quantitative results in Tables 1, 2, and 3 are computed by the released code (`eval_substitution.py`, `eval_ner_utility.py`) on the released artefact and are reproducible by any reader; no result was generated, projected, or hallucinated by the language model. The author is solely responsible for the technical claims, methodology, statistical analysis, and the final wording of the manuscript.

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A Locale-conditioned demonstration pools

For full reproducibility we list the demonstration pools used by the locale-conditioned rotating few-shot strategy (Section 3.2). Per-input MD5 hashes seed a deterministic sample of three demonstrations from the appropriate pool.

PERSON pool.

- `en`: (John Carter, Marcus Chen), (Linda Vasquez, Olivia Brennan), (David Kim, Theo Pemberton), (Sarah Patel, Maya Iyer), (Robert Williams, Daniel Foster), (Priya Krishnamurthy, Nadia Subramanian), (Michael O'Brien, Patrick Donovan), (Jennifer Wong, Cynthia Park).
- `de`: (Hans Müller, Karl Schmidt), (Anna Becker, Lena Hoffmann), (Klaus Wagner, Erik Krüger), (Ingrid Weber, Petra Neumann), (Stefan Fischer, Dietrich Bauer), (Helga Zimmermann, Brigitte Klein).
- `es`: (Juan García, Carlos Hernández), (María Rodríguez, Ana Fernández), (Diego Sánchez, Luis Castillo), (Carmen Ortiz, Lucía Vázquez), (Roberto Jiménez, Pablo Morales), (Sofía Ramírez, Elena Aguilar).
- `ja`: (山田太郎, 鈴木一郎), (佐藤花子, 田中美咲), (渡辺健, 高橋翔), (中村裕子, 小林由香), (加藤博之, 斎藤大輔), (井上恵美, 松本香織).
- `zh`: (李伟, 王芳), (张敏, 刘洋), (陈杰, 黄燕), (周磊, 吴娟), (徐明, 孙丽), (郑强, 马晶).

ADDRESS pool. Six entries per locale; en covers both en_US and en_IN. Examples include (`Hauptstraße 45, 10117 Berlin`, `Lindenallee 12, 80331 München`) for `de`, (`Calle Reforma 123, 06600 CDMX`, `Avenida Insurgentes 456, 03100 CDMX`) for `es`, and (北京市朝阳区建国路1号, 上海市浦东新区世纪大道100号) for `zh`. Full lists are in `pii_substitute.py`.

DATE pool. Pools per detected format: `mdy_slash` (5 entries), `ymd_dash` (4), `dmy_dash_mon` (4), `dmy_slash` (3), `unknown` (2 fallbacks).